Product Requirements Document (PRD)

CyberLabX – Virtual Cybersecurity Laboratory Platform

Version: 1.0

Product Owner: Devesh Shukla

Document Status: Draft

Last Updated: July 2026

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1. Product Overview

Product Name

CyberLabX

Product Vision

CyberLabX is a browser-based Virtual Cybersecurity Laboratory that enables students to perform real cybersecurity experiments in an isolated, interactive, and guided environment without installing complex software such as Kali Linux, Nmap, Wireshark, or Docker on their personal computers.

Instead of learning cybersecurity through PDFs and screenshots, students gain hands-on experience by launching virtual labs, executing commands, analyzing outputs, completing guided tasks, and receiving automated feedback and assessments.

CyberLabX aims to modernize cybersecurity education by making practical learning accessible, scalable, and engaging for educational institutions.

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2. Problem Statement

Traditional cybersecurity practical sessions face several challenges:

- Students struggle with installing and configuring cybersecurity tools.
- Different operating systems create inconsistent learning experiences.
- Faculty spend significant time troubleshooting installations instead of teaching concepts.
- Practical evaluations rely on screenshots and manual report checking.
- Students often copy commands without understanding the underlying concepts.
- Limited laboratory infrastructure restricts access to hands-on learning.

These issues reduce learning effectiveness and make cybersecurity education difficult to scale.

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3. Solution

CyberLabX provides each student with an isolated cybersecurity lab accessible directly from a web browser.

Students can:

- Launch experiments instantly.
- Use a browser-based Linux terminal.
- Perform real cybersecurity exercises.
- Receive interactive guidance.
- Understand networking through visual explanations.
- Be evaluated automatically.
- Generate experiment reports instantly.

Faculty can:

- Monitor student progress.
- Assign experiments.
- View reports.
- Export assessment results.

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4. Target Users

Primary Users

Undergraduate Students

- B.Tech
- BCA
- MCA
- Diploma

Learning cybersecurity fundamentals.

Faculty

Teaching

- Ethical Hacking
- Network Security
- Information Security
- Computer Networks

Educational Institutions

Engineering colleges

Universities

Training institutes

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5. User Personas

Student

Name

Rahul

Age

20

Goals

- Learn ethical hacking
- Complete lab experiments
- Prepare for placements
- Gain practical skills

Pain Points

- Installation problems
- Command confusion
- Lack of guidance
- Manual report writing

Faculty

Name

Dr. Sharma

Goals

- Conduct practical sessions
- Track student performance
- Reduce manual grading
- Standardize experiments

Pain Points

- Students have different environments
- Time wasted fixing installations
- Manual report verification

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6. Product Goals

Educational Goals

- Improve practical cybersecurity learning.

- Provide standardized lab environments.
- Encourage experimentation without risk.

Technical Goals

- Browser-based labs.
- Containerized infrastructure.
- Automatic grading.
- Secure isolated environments.

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7. Core Value Proposition

CyberLabX transforms static cybersecurity practical manuals into fully interactive virtual laboratories where students perform real experiments instead of merely reading instructions.

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8. Features

Must-Have Features (MVP)

User Authentication

- Student login
- Faculty login
- JWT authentication
- Password reset

Dashboard

Student dashboard showing

- Available experiments
- Completed experiments
- Progress

- Scores

Experiment Catalog

List of experiments

Initially:

- Network Mapping using Nmap

Theory Module

Each experiment includes

- Aim
- Objectives
- Theory
- Procedure
- Safety instructions

Interactive Lab

Students can

- Launch a virtual environment
- Access a Linux terminal
- Execute commands
- View outputs

Browser Terminal

Embedded Linux terminal supporting

- Nmap
- Tcpdump
- Basic Linux commands

Guided Tasks

Example

Task 1

Find live hosts

Task 2

Perform SYN Scan

Task 3

Identify OS

Task 4

Detect Services

Automatic Evaluation

System verifies

- Commands executed
- Correct sequence
- Task completion

Generates marks automatically.

Quiz

MCQ assessment after experiment completion.

Report Generation

Generate downloadable PDF containing

- Student details
- Commands executed
- Outputs

- Marks
- Faculty remarks

Faculty Dashboard

Faculty can

- View students
- View reports
- View scores
- Assign experiments

Admin Panel

Manage

- Students
- Faculty
- Experiments

Nice-to-Have Features

Network Packet Visualization

Animated packet flow while commands execute.

AI Learning Assistant

Context-aware assistant that provides hints without revealing complete answers.

Interactive Network Topology

Visual network map showing

- Hosts

- Connections
- Open ports

Real-Time Collaboration

Instructor can join student labs.

Voice Explanations

Explain commands verbally.

Leaderboards

Gamified learning.

Certificates

Automatic certificate generation.

Dark/Light Theme

User preference.

Experiment Replay

Replay previous sessions.

Mobile Companion App

Progress tracking.

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9. User Journey

Student Flow

```
None
Landing Page

↓

Login

↓

Dashboard

↓

Choose Experiment

↓

Read Theory

↓

Launch Virtual Lab

↓

Browser Terminal Opens

↓

Complete Guided Tasks

↓
```

```
Automatic Evaluation

↓

Quiz

↓

Report Generated

↓

Dashboard Updated
```

Faculty Flow

```
None
Login

↓

Faculty Dashboard

↓

Assign Experiment

↓

Monitor Progress

↓

Review Reports
```

```
↓

Export Results
```

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10. MVP Scope

Version One includes only one fully functional cybersecurity experiment.

Experiment:

Network Mapping using Nmap

Students should be able to

- Launch isolated lab
- Execute Nmap commands
- Understand outputs
- Complete guided exercises
- Take quiz
- Download report

No other cybersecurity experiments will be included initially.

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11. Success Metrics

Student Metrics

- Experiment completion rate
- Quiz scores
- Time taken
- Number of retries
- Student satisfaction

Faculty Metrics

- Reduction in grading time
- Number of experiments conducted
- Student participation

Platform Metrics

- Daily active users
- Weekly active users
- Average session duration
- Number of completed labs
- Report downloads

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12. Non-Goals (Version One)

The following features are intentionally excluded from the first release.

No Offensive Security Against Public Systems

Students cannot scan

- Public IPs
- External servers
- College infrastructure

Only isolated lab environments.

No Capture The Flag (CTF)

Competitive hacking is postponed.

No Multiplayer Labs

One student per environment.

No AI Tutor

Planned for Version Two.

No Wireshark Experiment

Future release.

No SQL Injection Lab

Future release.

No Password Cracking

Future release.

No Mobile Application

Web only.

No Cloud Deployment

Initially hosted on a local server or college infrastructure.

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13. Functional Requirements

Students must be able to

- Register/Login
- Launch experiments
- Execute Linux commands
- Complete tasks
- View results
- Download reports

Faculty must be able to

- Assign experiments
- Monitor students
- View reports

Admins must be able to

- Manage users
- Manage experiments
- View system health

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14. Non-Functional Requirements

- Browser-based
- Responsive UI
- Secure authentication
- Docker-based isolation
- Support 100+ concurrent users (target)
- Average lab startup time under 10 seconds
- HTTPS communication
- Role-based access control

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15. Risks

Risk	Mitigation

High server load	Use lightweight Docker containers instead of full VMs
Students attempting unauthorized scans	Restrict networking to an isolated lab network
Long container startup times	Pre-build container images and reuse templates
Docker security concerns	Run containers with restricted privileges and resource limits
Faculty unfamiliar with the platform	Provide onboarding guides and documentation

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16. Future Roadmap

Version 1.1

- Wireshark Lab
- Tcpdump Lab
- Linux Basics

Version 1.2

- SQL Injection
- XSS
- CSRF

Version 2.0

- Password Cracking
- Burp Suite
- Hydra
- Cryptography
- Digital Forensics

Version 3.0

- AI Tutor
- Adaptive Learning
- Packet Visualization
- Capture The Flag
- Multiplayer Labs
- LMS integration

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17. Product Summary

CyberLabX is an educational platform designed to replace traditional cybersecurity practical manuals with an interactive, browser-based laboratory experience. By combining isolated lab environments, guided exercises, automatic assessment, and faculty management tools, it addresses the major pain points of current cybersecurity education. The MVP focuses on delivering one complete, high-quality Nmap experiment rather than many incomplete labs, establishing a scalable foundation for future cybersecurity modules.